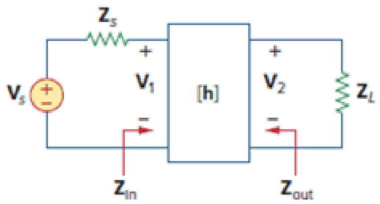
Problem

The *h* parameters of the two-port of Fig. 19.95 are:

$$[\mathbf{h}] = \begin{bmatrix} 600\ \Omega & 0.04 \\ 30 & 2\ \text{mS} \end{bmatrix}$$

Given the *Zs* = 2 kΩ and *ZL* = 400 Ω, find *Z*in and *Z*out.

Figure 19.95



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

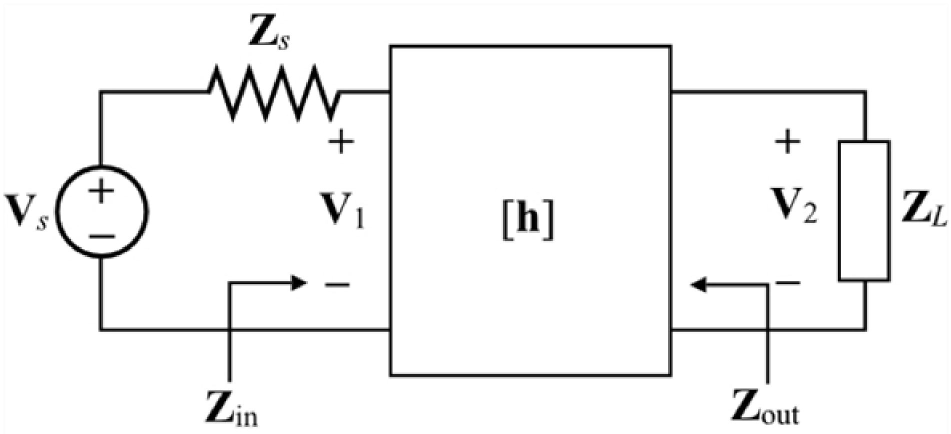


Figure 1

Step 2 of 4

Draw the equivalent circuit diagram of Figure 1.

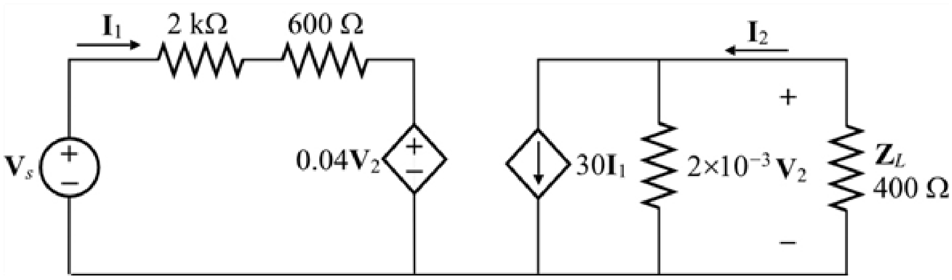


Figure 2

Step 3 of 4

Calculate the input impedance.

$$\begin{aligned}Z_{\text{in}} &= \mathbf{h}_{11} - \frac{\mathbf{h}_{12}\mathbf{h}_{21}R_L}{1+\mathbf{h}_{22}R_L}\\ &= 600 - \frac{(0.04)(30)(400)}{1+(0.002)(400)}\\ &= 600 - \frac{480}{1+0.8}\\ &= 600 - \frac{480}{1.8}\\ &= 600 - 266.67\\ &= 333.33\,\Omega\end{aligned}$$

Therefore, the input impedance, Z_{in} is $\boxed{333.33\,\Omega}$.

Step 4 of 4

Calculate the output impedance.

$$\begin{aligned}Z_{\text{out}} &= \frac{R_s + \mathbf{h}_{11}}{(R_s + \mathbf{h}_{11})\mathbf{h}_{22} - \mathbf{h}_{21}\mathbf{h}_{12}}\\ &= \frac{2000+600}{(2000+600)(0.002)-(30)(0.04)}\\ &= \frac{2600}{(2600)(0.002)-1.2}\\ &= \frac{2600}{5.2-1.2}\\ &= \frac{2600}{4}\\ &= 650\,\Omega\end{aligned}$$

Therefore, the output impedance, Z_{out} is $\boxed{650\,\Omega}$.